

Advanced Animal Movement Modelling Training Workshop

Species-Habitat Associations

17 September 2026

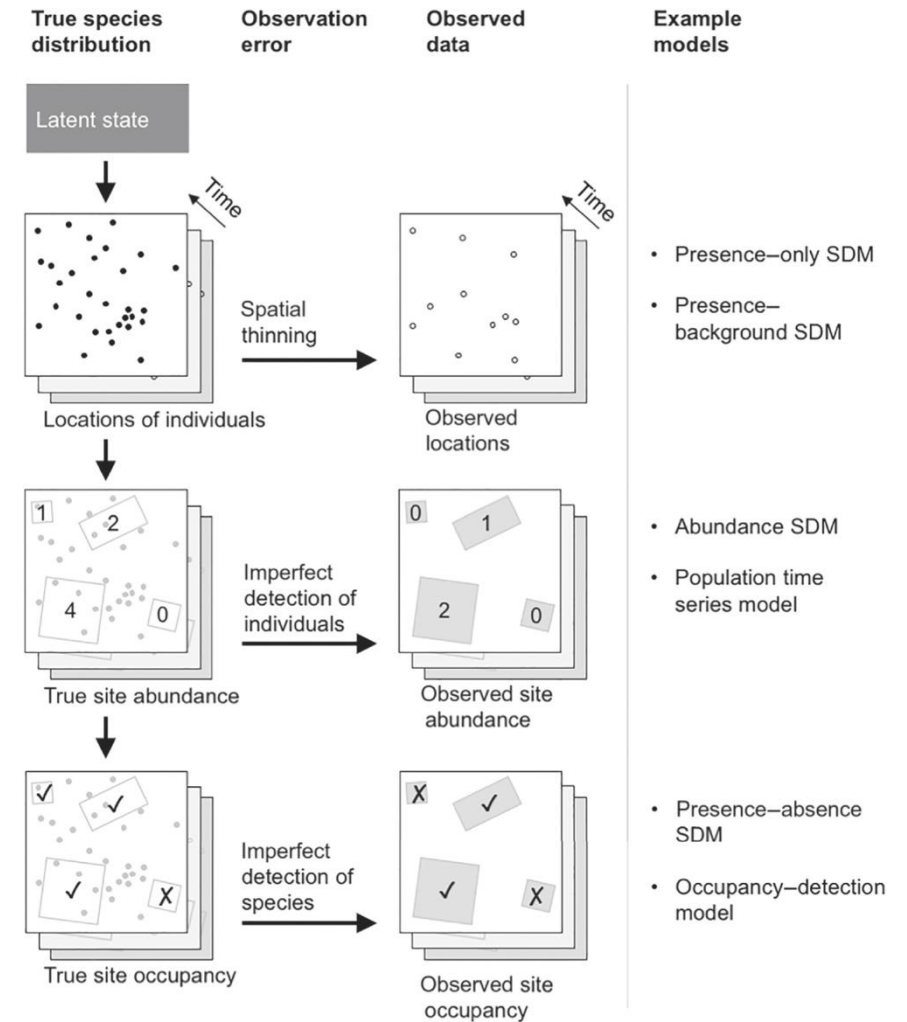
Josh Cullen



Photo credit: Larry Li

What are species-habitat associations?

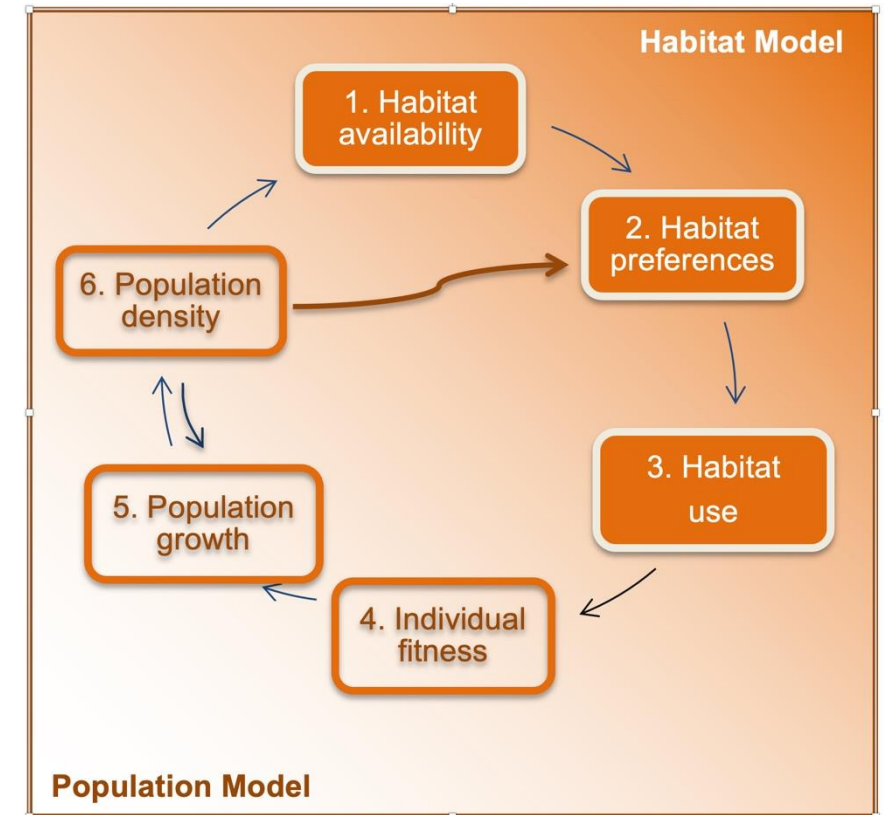
- Per Matthiopoulos et al. (2023):
 - “SHA connects aspects of habitat (resources, conditions, risks) to particular observations about a species (recorded at the individual, group or population levels).”
 - “Species Distribution Models (SDMs) are the foremost available batch of methods that achieve this statistical connection.”



Primary assumptions of SDMs

- Species only occur at suitable habitat and aren't found in unsuitable habitat
 - Related to fundamental niche
- Higher suitability habitat associated with higher species density and fitness

These assumptions are often violated related to density-dependence, behavioral or ontogenetic differences in habitat needs, other drivers (e.g., predator-prey dynamics, mismatch in timing of phenomenon, etc)



Common types of animal data for SDMs

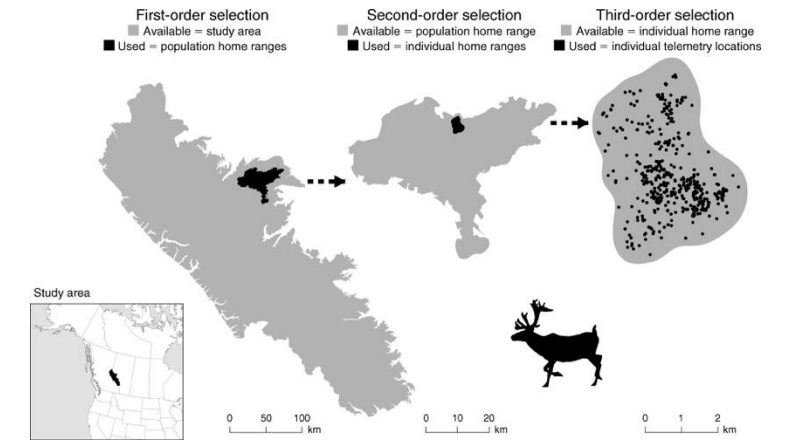
- While we'll focus on telemetry data, here are other examples of data types that can be used:
 - **Abundance:** point counts, transect counts, plot counts, camera traps
 - **Presence-absence:** surveys (point/transect/plot), mark-recapture, eDNA, camera traps
 - **Presence-only:** telemetry, community science data, museum specimens

SDM model types:

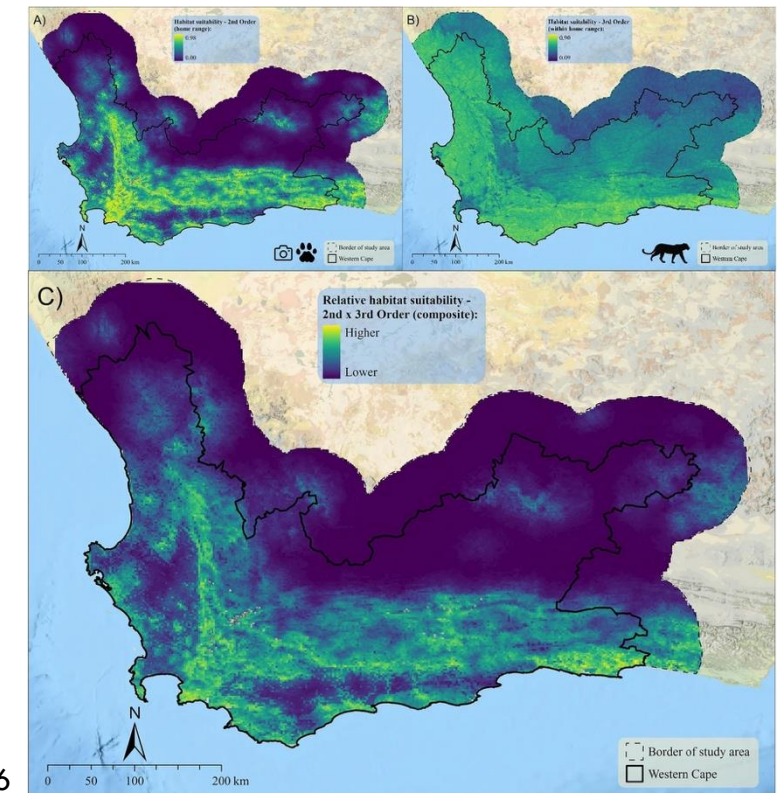
- **Occupancy models**
- **Distance sampling models**
- **Density surface models**
- **N-mixture models**
- **Habitat selection models**
- **Habitat suitability models**

Scales of habitat selection

- 3 orders of habitat selection per Johnson (1980)
- Estimates population-level selection
- The order of selection evaluated depends on study goals
- The order of selection also dictates where “available” or “background” points are generated for presence-only telemetry data



DeCesare et al. 2012

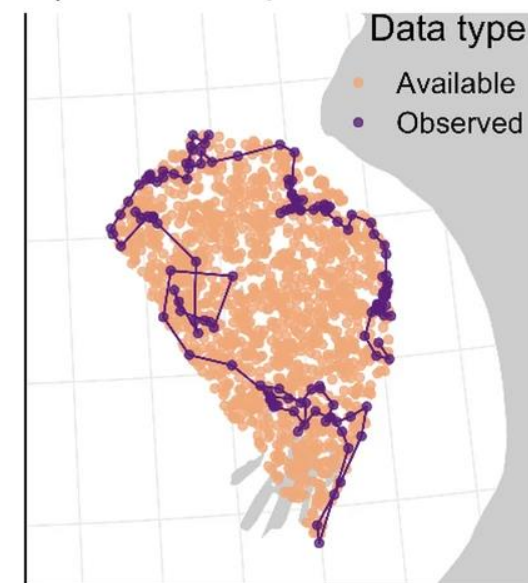


Greyling et al. 2026

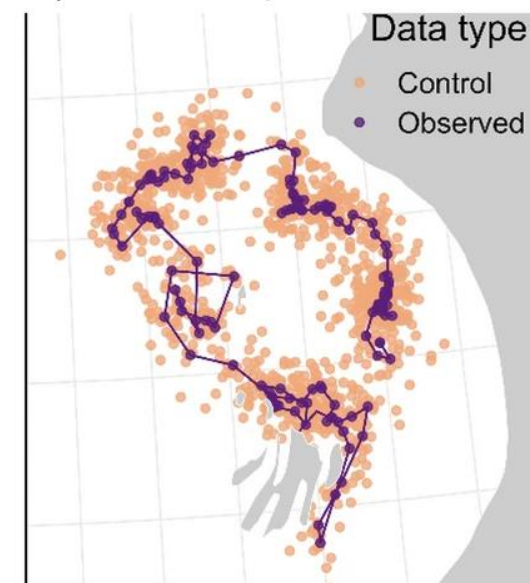
Types of selection models

- Main 2 are resource selection functions (RSFs) and step selection functions (SSFs)
 - **RSFs**: “available” points generated throughout at least individual’s home range if not further
 - **SSFs**: estimate selection for each step of movement process, not the animal’s range
- Model choice depends on process of interest
- Possible to do multi-scale approach by fitting both

A) RSF samples

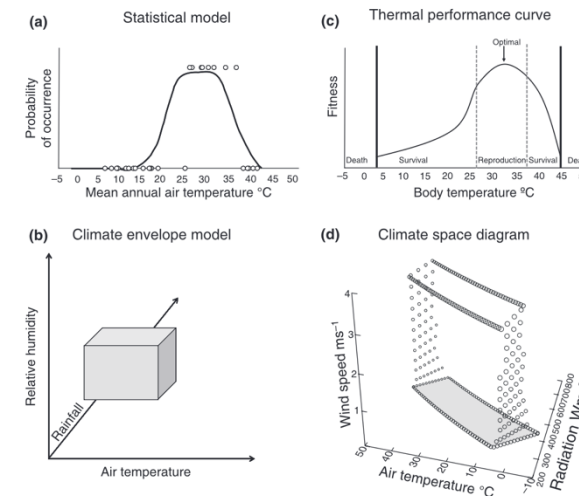
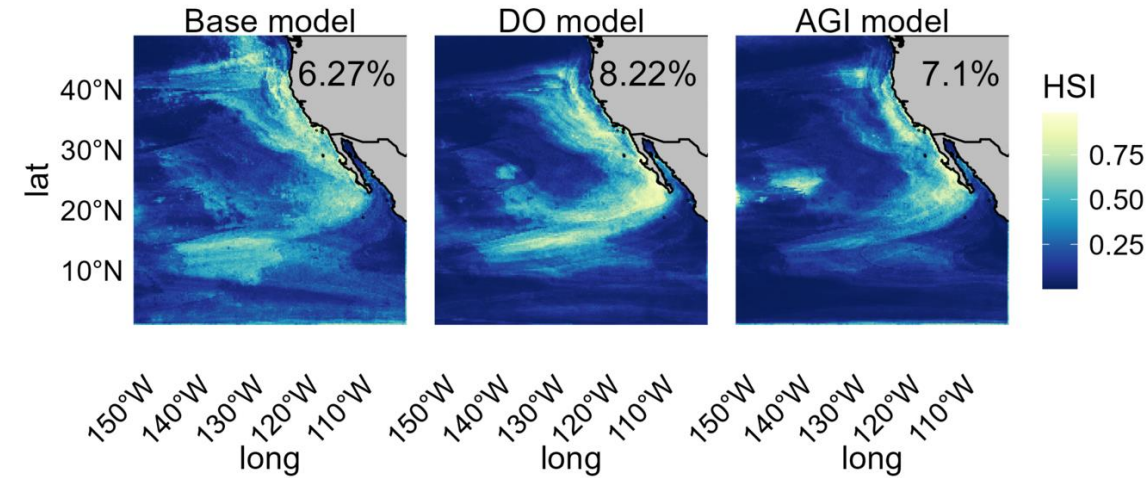


B) SSF samples



More than just correlative models...

- In general, SDMs often fitted as correlative models
 - i.e., distribution associated with habitat features
- Mechanistic models take the opposite approach by applying knowledge of species decision-making, physiology, dispersal kernel, etc
 - Often through IBMs/ABMs
- Some models alternatively fitted as hybrid that incorporate some knowledge of species into standard SDM



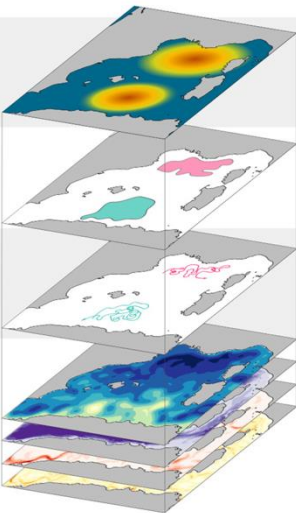
Nazario et al. 2025

Kearney & Porter 2009

Bridging individual- to population-level processes

- Telemetry data inherently include information about individual-level movements, but this can also be used to make population-level inferences
- Other datasets, such as surveys and forms of unmarked sampling, inherently sample at the population level

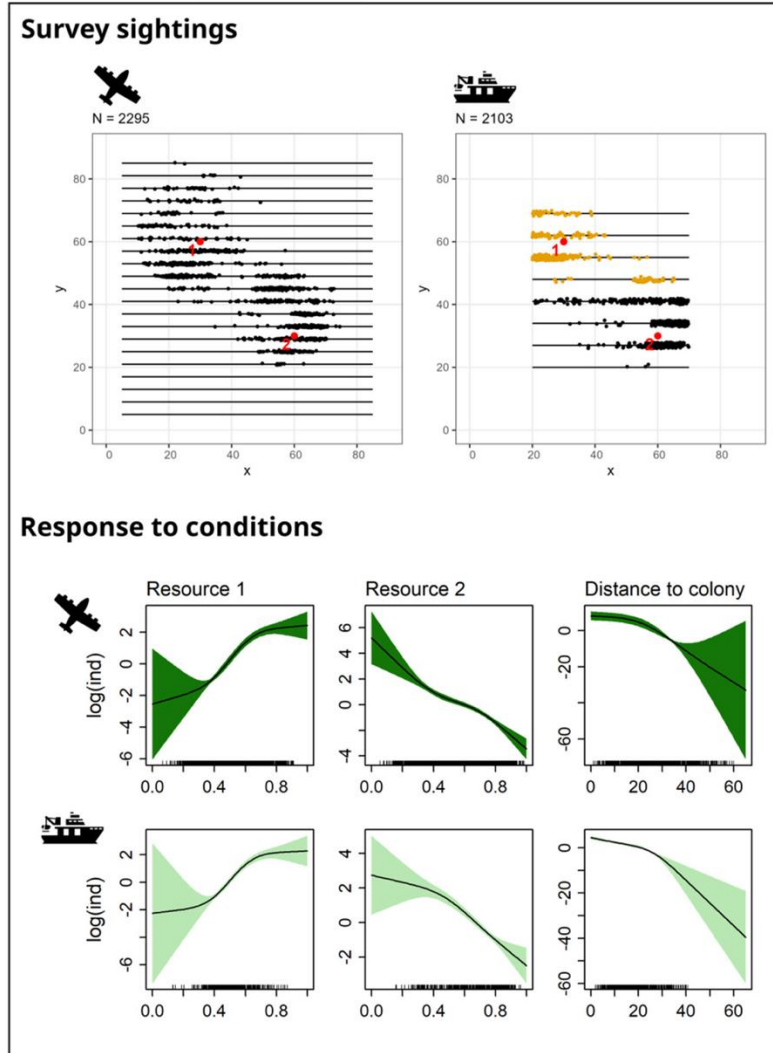
A - From individual- to population-scale processes

Scale	Process		Data	SDM type	Metric
Population	Abundance Occurrence		Observation	Presence-absence Presence-background Count	Presence probability Use probability Abundance
	Home range		Movement		Utilisation distribution
Individual	Movement Habitat choice		Movement	Presence-background	Use probability
	Environmental conditions				

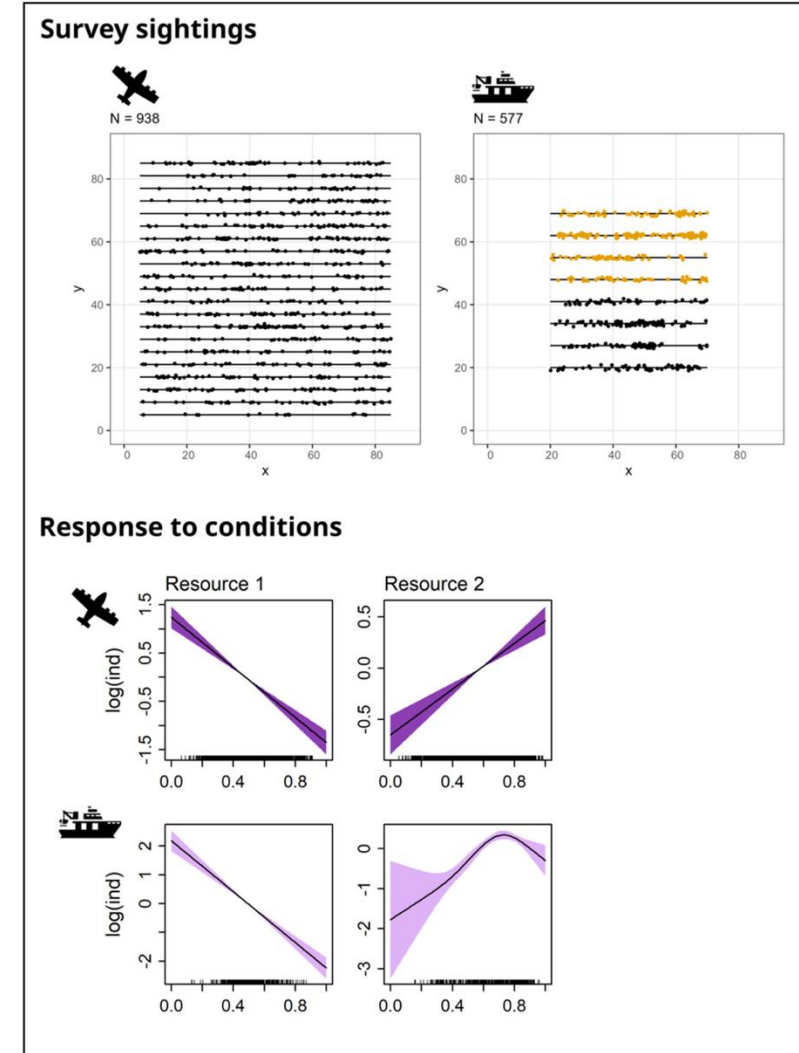
Bridging individual- to population-level processes

- Differences likely arise from interaction between sampling bias and animal movement patterns

A - Central-place forager



B - Free ranger



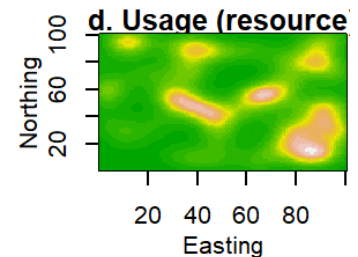
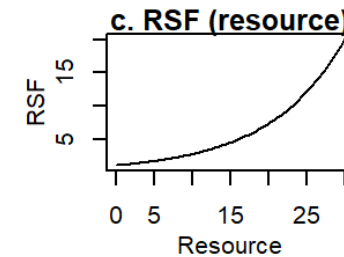
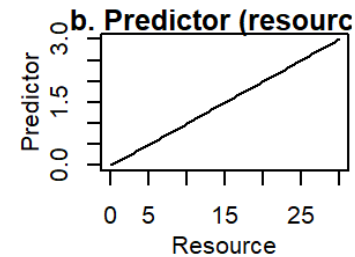
Resource selection functions (RSFs)

- Technically, a component in the numerator of expression to estimate habitat use
- Proportional to habitat preference when intercept omitted
- Often fitted using form of logistic regression
 - But exponentiating linear predictor instead of using inverse logit link
- Under certain conditions, approximates an inhomogeneous Poisson point-process (IPP)

$$g(\mathbf{x}) = \sum_{k=1}^K a_k x_k$$

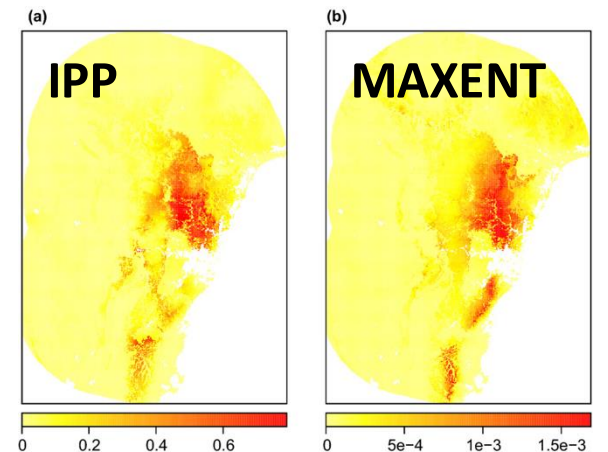
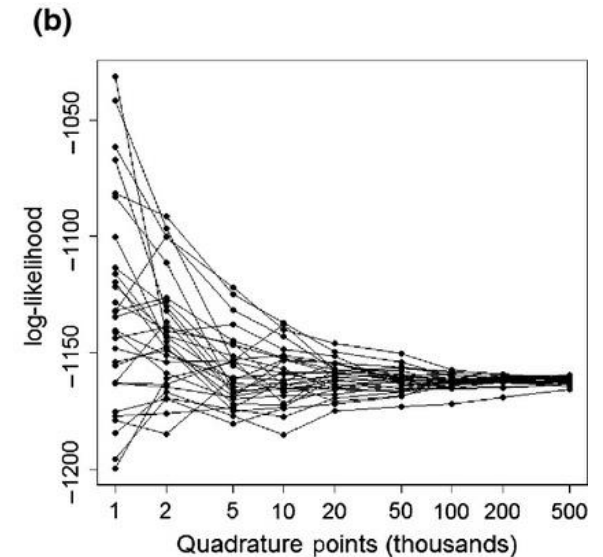
RSF $\rightarrow w(\mathbf{x}) = \exp(g(\mathbf{x}))$

$\rightarrow f_u(\mathbf{x}) = \frac{w(\mathbf{x}) f_a(\mathbf{x})}{\int_{\mathbb{E}} w(\mathbf{z}) f_a(\mathbf{z}) d\mathbf{z}}$



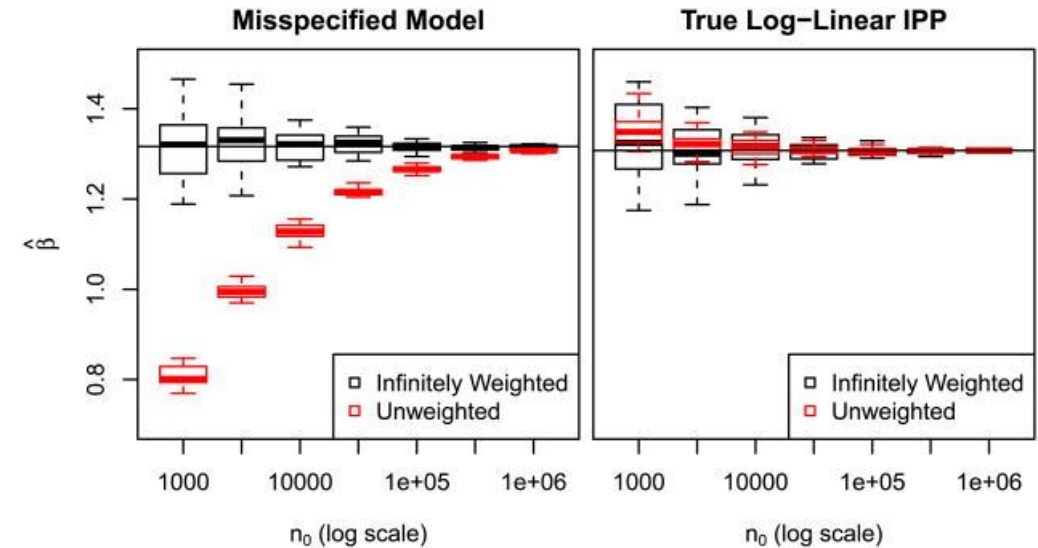
Connection between presence-only models and inhomogeneous Poisson process (IPP)

- RSF SDMs have been shown to approximate an IPP via asymptotic equivalence (Warton & Shepherd, 2010)
 - Essentially translating SDMs from environmental (E) space to geographic (G) space
 - Spatial point intensity $[\lambda(s)]$ and RSF $[w(x)]$ are proportional
- To achieve this with RSFs, you either need large number of “available” points or to use weights on likelihood



Types of weighted RSFs

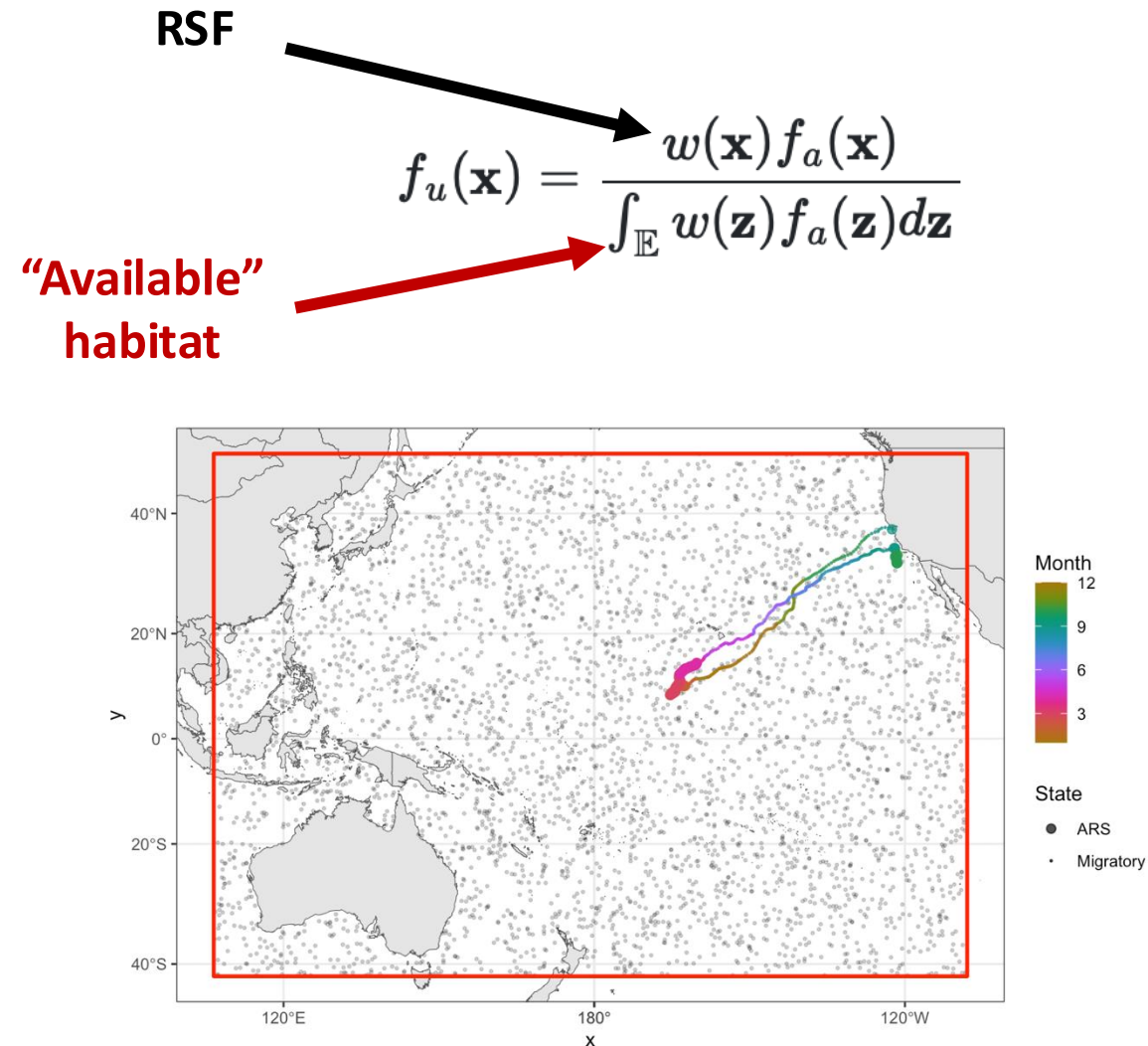
- Infinitely weighted logistic regression (IWLR; Fithian & Hastie, 2013)
 - $Y = 1$ (weight = 1); $Y = 0$ (weight = 5000)
- Down-weighted Poisson regression (DWPR; Renner et al., 2015)
 - $Y = 1$ (weight = $1e-6$); $Y = 0$ (weight = Area sampled / N_{avail})
 - For this model, response (Y) is also divided by weights when fitting model



Fithian & Hastie, 2013

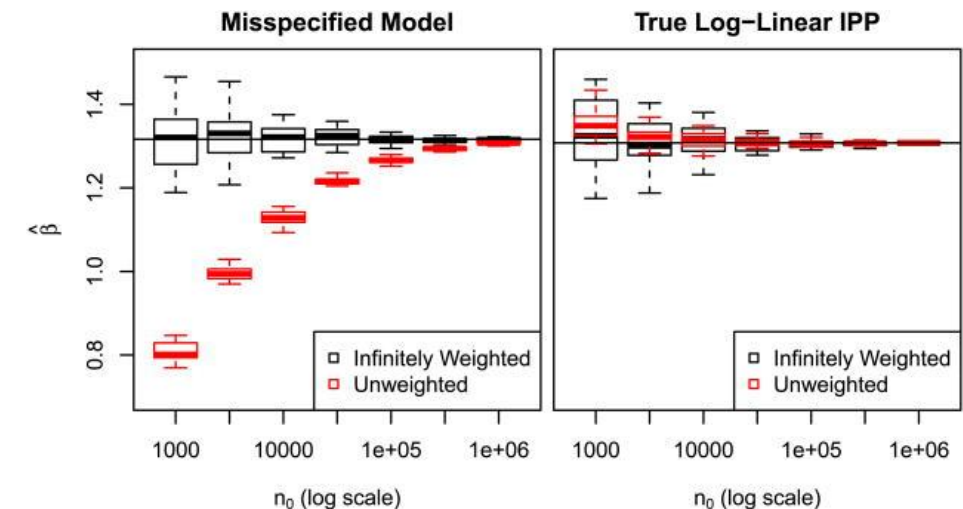
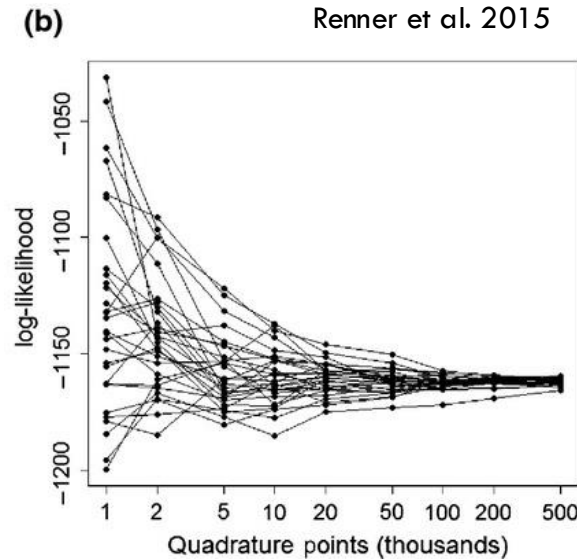
“Available” points as a form of numerical integration

- While the order of habitat selection may dictate **where** ‘available’ points are generated, this doesn’t dictate **how**
- “Available” points used to approximate integral in denominator of habitat use expression
- Points typically generated through Monte Carlo integration (i.e., large number of random “background” points)
- Can also use numerical integration via use of quadrature points (e.g., points covering each pixel of the landscape)



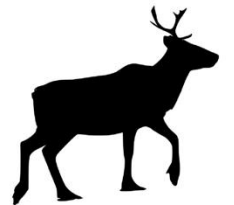
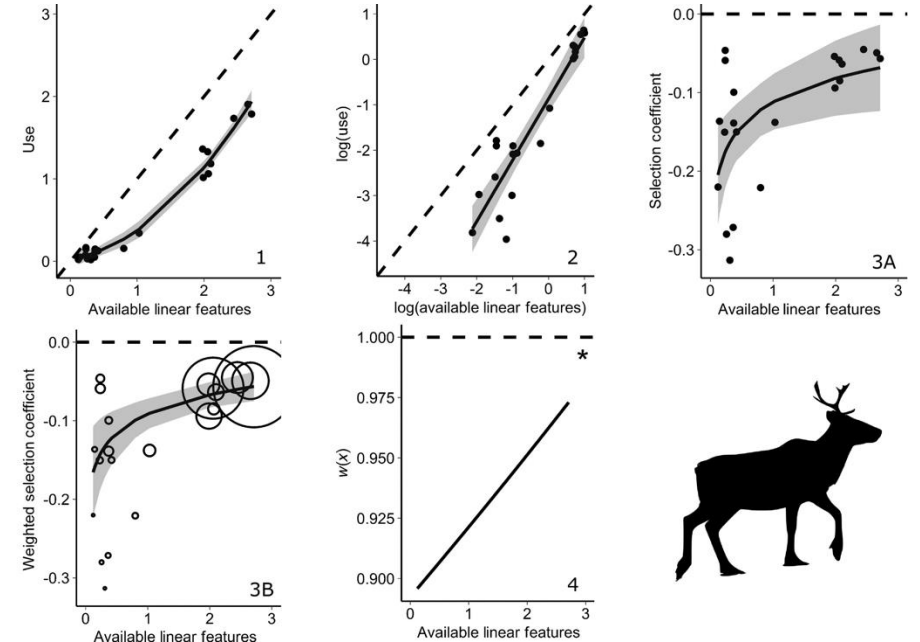
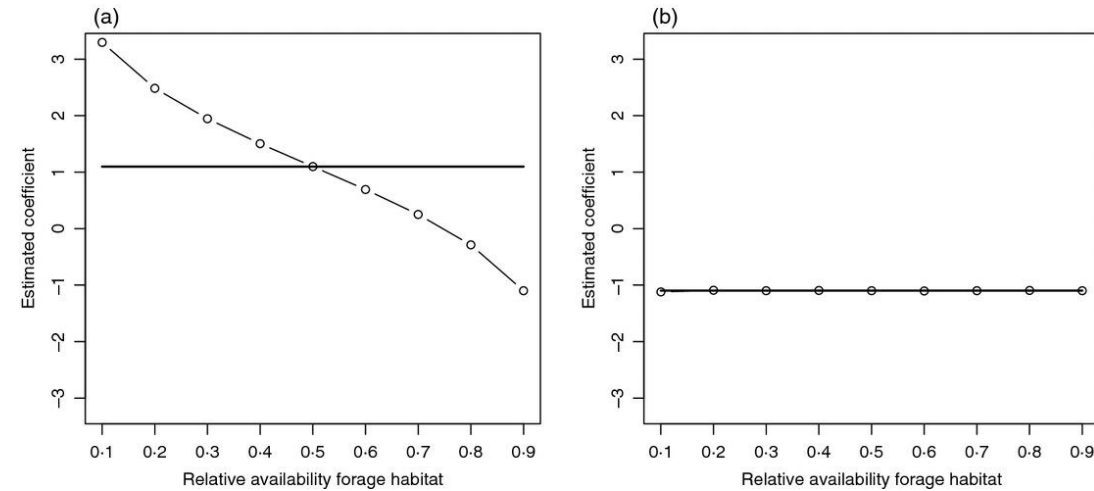
“Available” points as a form of numerical integration

- There’s no exact recommended number of available points to use, since this will also vary by weights used and “roughness” of the intensity surface (Fieberg et al. 2021; Renner et al. 2015)
- General advice is to fit models with increasing number of available points until selection coefficients no longer change much (Fieberg et al. 2021)
 - E.g., can try ratios of $N_{\text{avail}}:N_{\text{used}}$ of 10:1, 30:1, 50:1 and increase if necessary



Functional responses

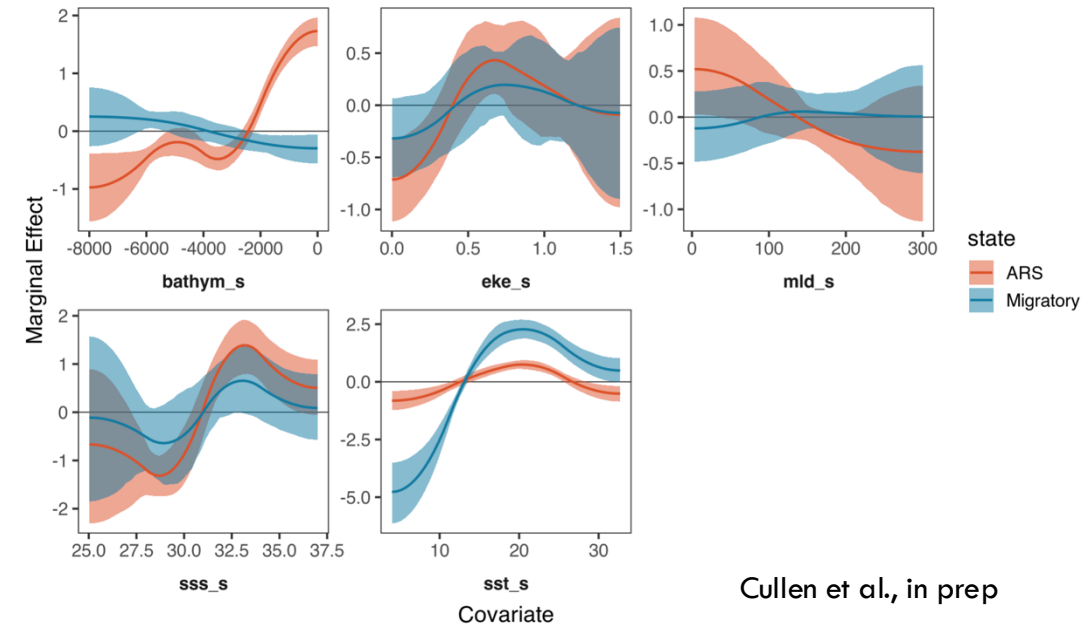
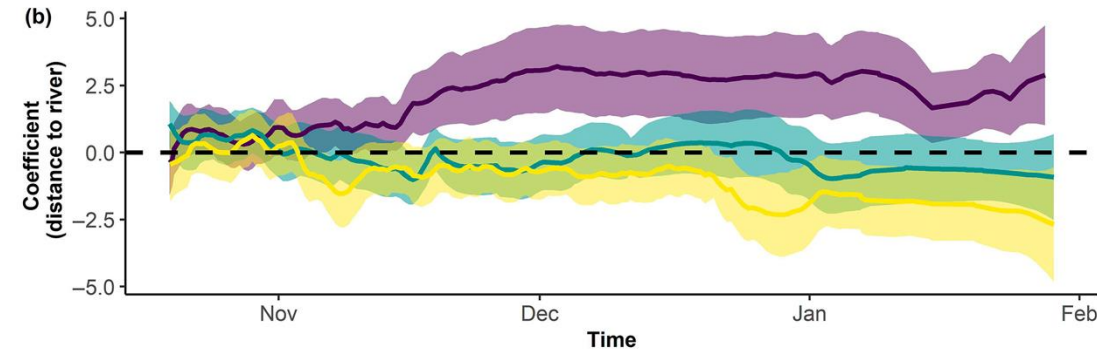
- Animals often modify their selection of habitat depending on habitat context (Mysterud & Ims, 1998; Matthiopoulos et al. 2023)
 - Due to density-dependent relationships
 - Due to physiological or survival tradeoffs
 - Due to non-linear (and possibly asymptotic) relationships of selection given availability
- Can potentially account for this via generalized functional responses (Matthiopoulos et al. 2011)



Extensions of RSFs

- Many ways that RSFs can be modified and applied to add complexity:
 - Use of ID-level random effects
 - Inclusion of spatial (and/or temporal) random effect to account for autocorrelation
 - Joint SDMs (JSDM) for multispecies modeling
 - Integrated SDMs (iSDM) to include multiple data types
 - Behavior-specific habitat selection
 - Time-varying habitat selection

Dejeante et al. 2024



Cullen et al., in prep

Step selection functions (SSFs)

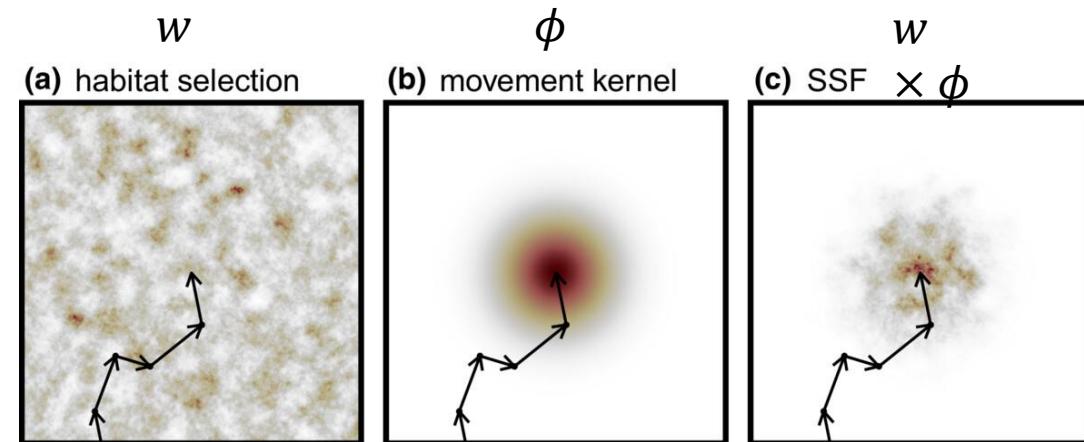
- Almost identical mathematical expression to RSF, but evaluated per step of movement
 - Unlike RSF, evaluates habitat selection while accounting for movement process
- Basic SSF (where “available” steps are drawn from empirical distributions) can be considered special case of iSSF (Michelot et al. 2024)
 - Where iSSF explicitly estimates movement kernel (of steps and turns)

$$\text{SSF} = w \times \phi$$

Michelot et al. 2024

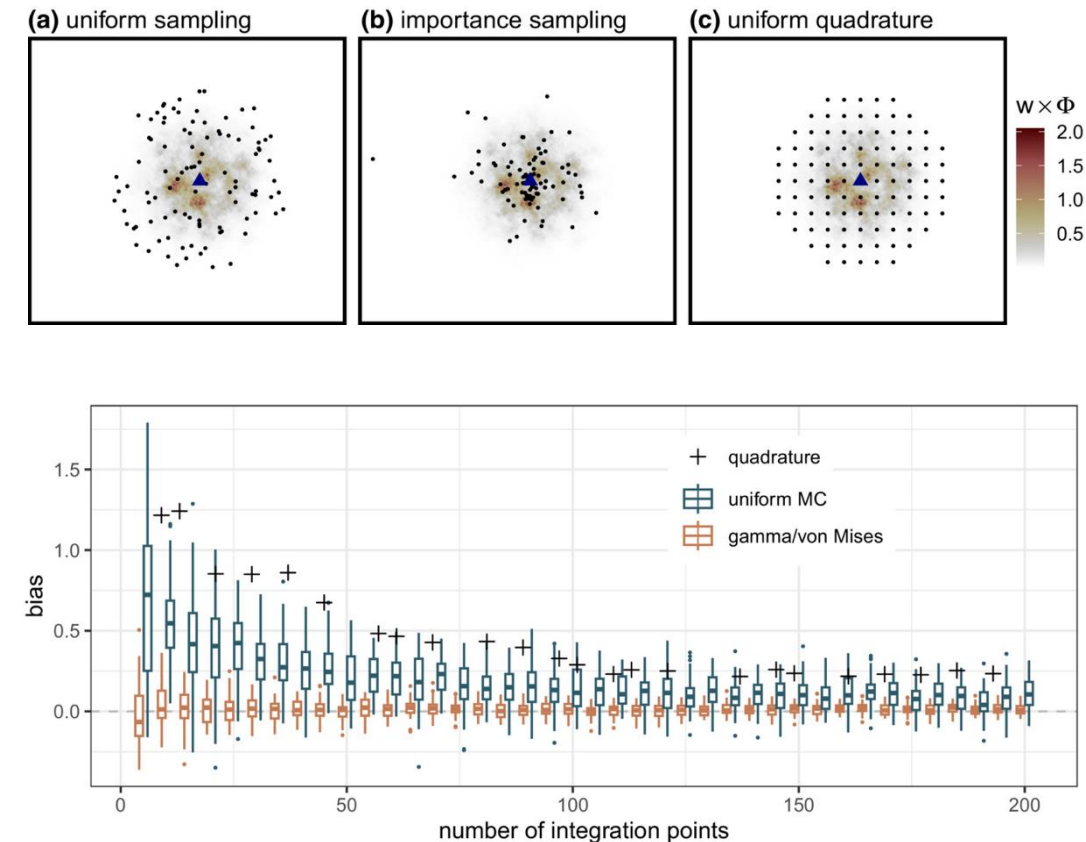
$$p(\mathbf{x}_{t+1} | \mathbf{x}_{1:t}) = \frac{w(\mathbf{x}_t, \mathbf{x}_{t+1}) \phi(\mathbf{x}_{t+1} | \mathbf{x}_{1:t})}{\int_{\Omega} w(\mathbf{x}_t, \mathbf{z}) \phi(\mathbf{z} | \mathbf{x}_{1:t}) d\mathbf{z}}$$

“Available” habitat



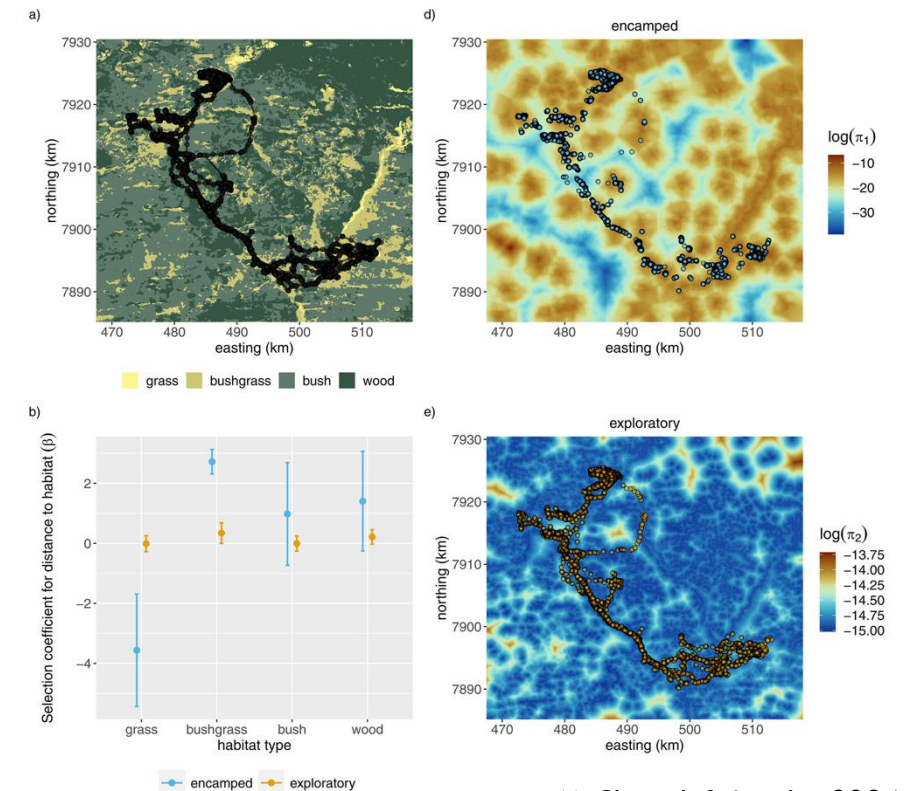
Distinguishing SSFs from statistical estimators

- SSFs are typically framed as “case-control” analysis using conditional logistic regression
 - Where each “used” step evaluated against “available” steps for time t
- This original framework can be loosened to allow fitting SSF in different ways
- “Available” steps just serve as a mathematical way to approximate integral (Michelot et al. 2024)
- As for RSFs, practitioners should test different number of available steps to asymptotically estimate true response (Michelot et al. 2024)



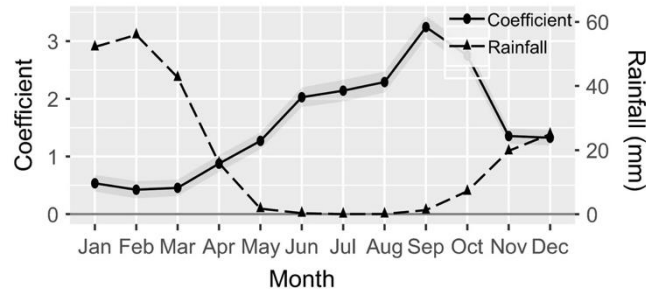
Extensions of SSFs

- Many ways to expand on (i)SSFs:
 - Use of ID-level random effects
 - HMM-SSF
 - Multistate Langevin diffusion
 - Temporally dynamic SSFs
 - Use of circular-linear copula distribution for movement kernel

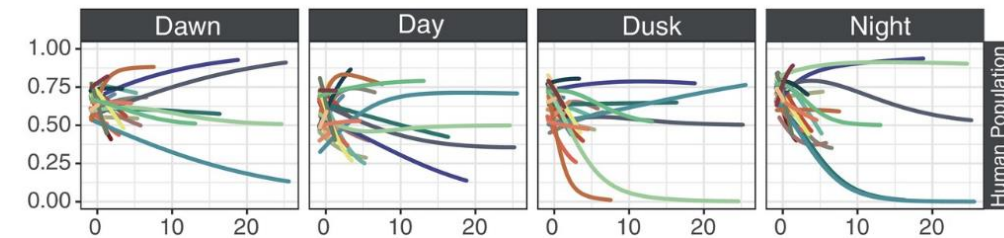
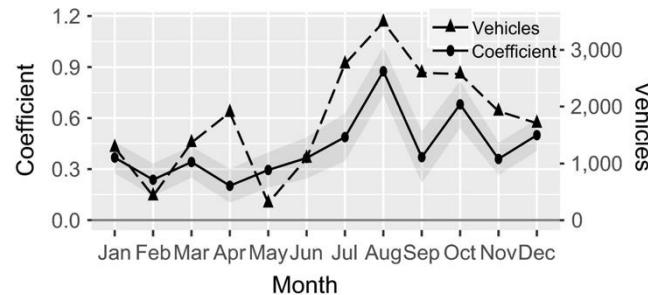


McClintock & Lander 2024

G) Distance to water



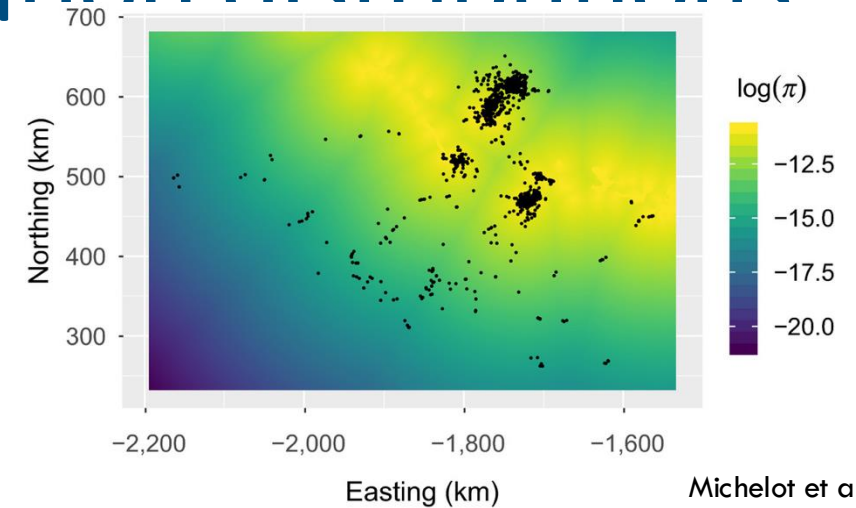
H) Distance to roads



Ferreiro-Arias et al. 2024

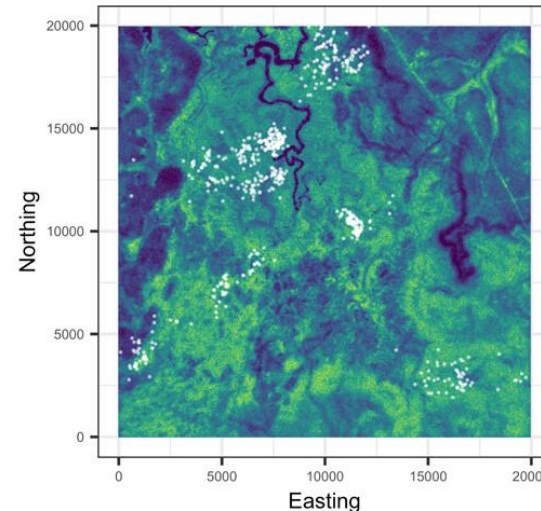
Linking RSFs and SSFs to utilization distributions

- Spatial predictions from RSF (via IPP) can be thought of as representing a UD
 - From “space use” perspective, can likely be thought of as ‘occurrence distribution’ (Matthiopoulos et al. 2023)
- Simulations from SSFs can be simulated to generate predictions of transient (if over short-term) or steady-state (over long-term) UD

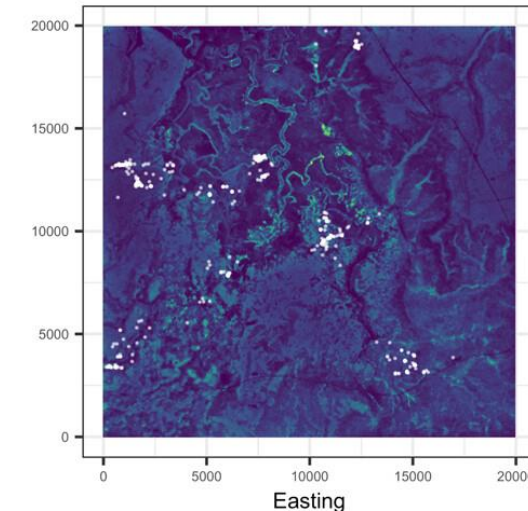


Michelot et al. 2019

(A) Model 2p - Hour 7



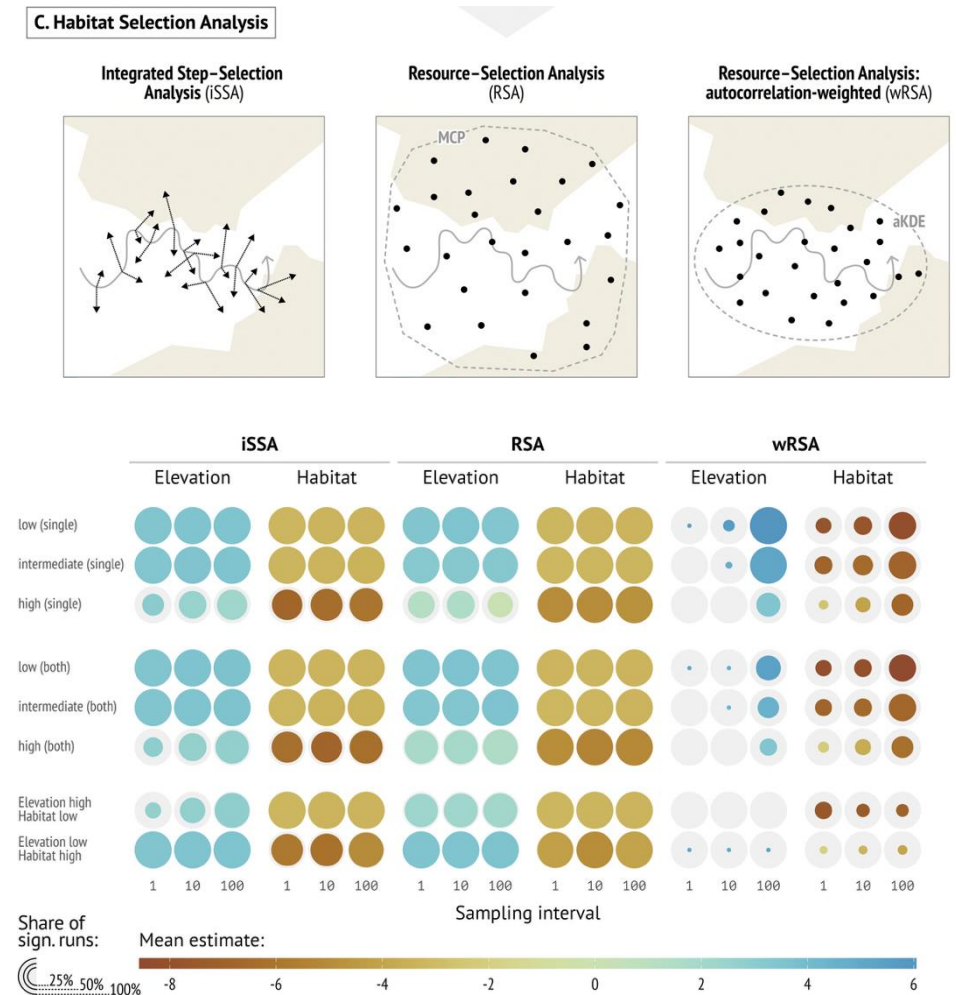
(B) Model 2p - Hour 12



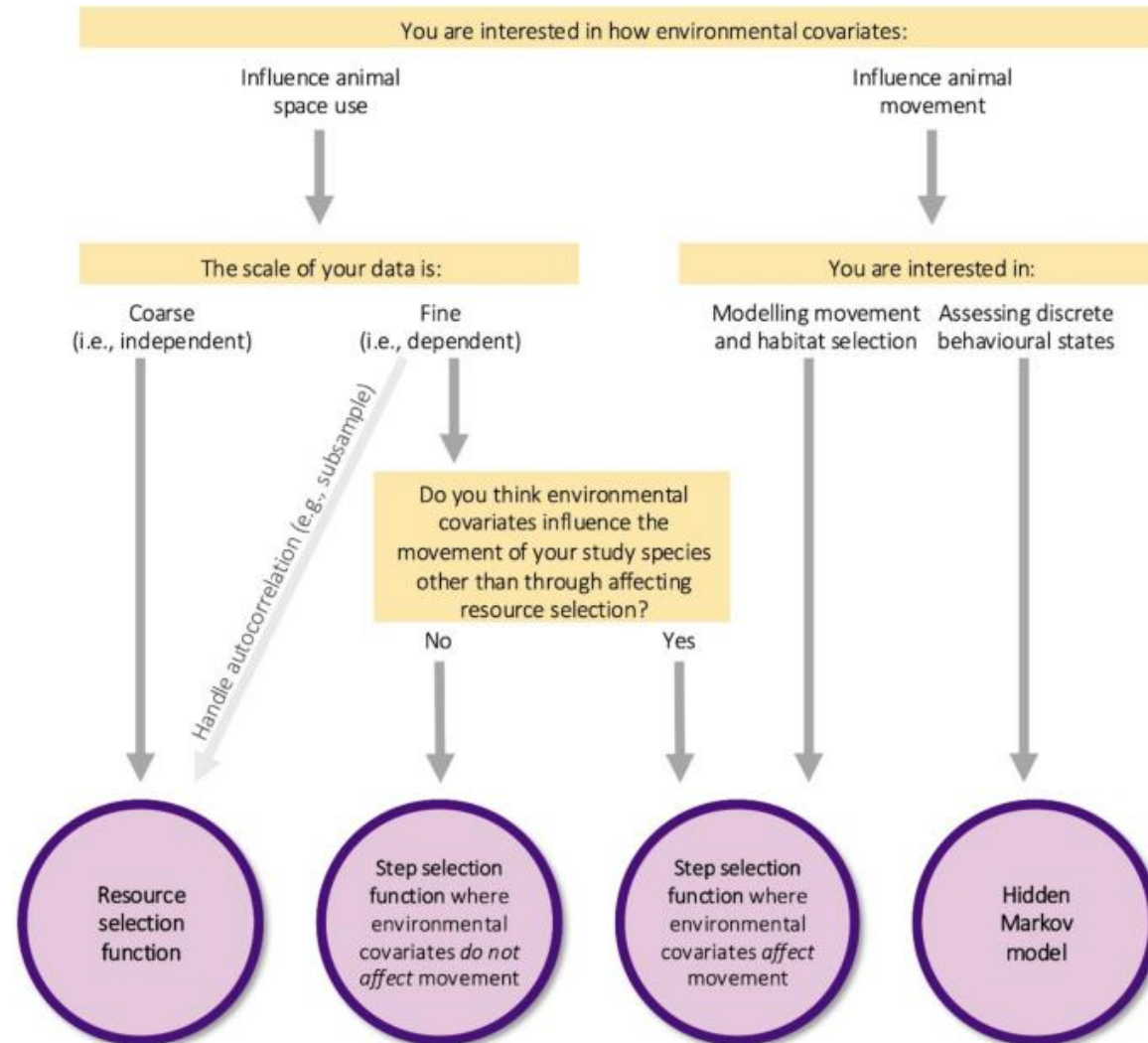
Forrest et al. 2025

Effect of temporal resolution and landscape configuration (Signer et al. 2025)

- Low to no effect of sampling interval on selection if no spatial autocorrelation on landscape
- RSF and iSSF more likely to estimate significant selection coefficients when spatial autocorrelation was low or intermediate
- Weighted RSF (by temporal autocorrelation in tracks) was less certain of significant effects overall



Choosing a habitat selection model



Time for live-coding



Fit RSF, SSF, and iSSF; visualize results, make inferences, and generate spatial predictions